

ICT - Semester 7

NATURAL LANGUAGE PROCESSING (20IC403T)

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Reference Books (Theory):

- Jurafsky, Daniel, and James H. Martin, Speech and Language Processing: An Introduction to Natural Language Processing, Speech Recognition, and Computational Linguistics, Prentice Hall. 2026 Edition.
- Yoav Goldberg, Graeme Hirst. Neural Network Methods in Natural Language Processing, Morgan & Claypool Publishers.
- Kamath, Uday, et al. Large Language Models: A Deep Dive.

Reference Book (Hands-on Practical):

- Bird, Steven, Ewan Klein, and Edward Loper. Natural language processing with Python: analyzing text with the natural language toolkit. O'Reilly Media, Inc.
- Thanaki, Jalaj. Python natural language processing. Packt Publishing Ltd.
- Alammam, J., & Grootendorst, M. Hands-On Large Language Models, O'Reilly.

Online Video Course and Support Materials:

- **NPTEL Course: Deep Learning for Natural Language Processing, IIT Kharagpur, Prof. Pawan Goyal**
 - <https://nptel.ac.in/courses/108105572>
 - <https://sites.google.com/view/dl4nlp-nptel/home>

NATURAL LANGUAGE PROCESSING

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UNIT 2

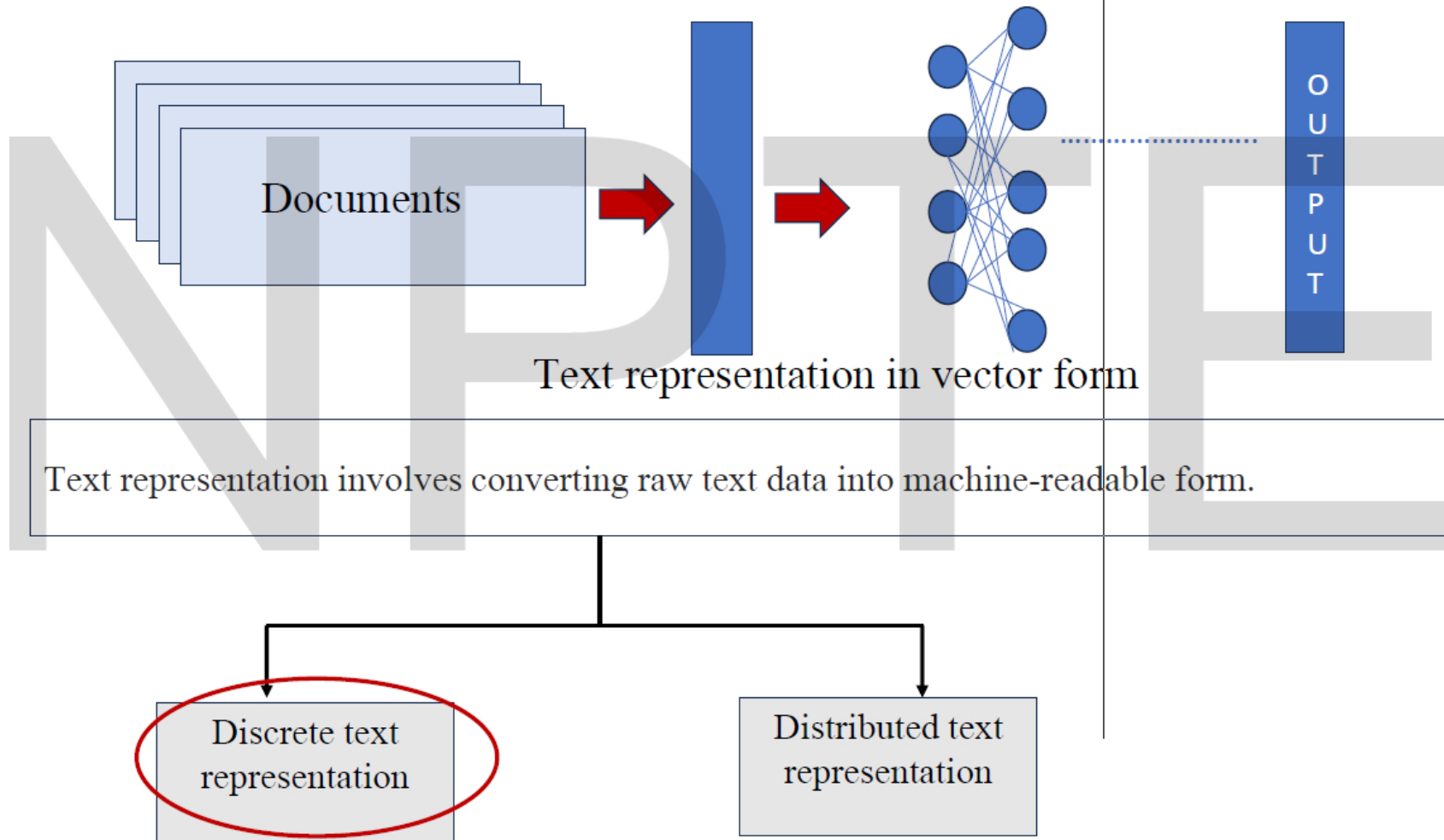
FEATURE REPRESENTATIONS AND EXTRACTIONS IN NLP

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Topics to cover:

- One-Hot Encoding
- Bag of Words
- TF-IDF Vectorization
- N-gram Language Models
- Word2Vec Embeddings
- Glove Embeddings
- FastText Embedding

Text Representation in NLP



- Text representation is a technique used in NLP to transform raw text into numerical features that capture information needed for computational processing and learning.

Text Representation in NLP

- Discrete Text Representation:
 - One Hot Encoding
 - Bag of words
 - TF-IDF (Term Frequency – Inverse Document Frequency)
- Distributed word representation
 - Word2Vec
 - Skip Gram model
 - Continuous Bag Of Words model
 - Word Embeddings in NLP
 - GloVe Embedding
 - FastText Embedding

One-Hot Encoding

One Hot Encoding is a method for converting **categorical variables** into a binary format.

↓
Representing qualities and characteristics

It changes every word to a $|V|$ dimensional vector.

Categorical Variable
Red
Blue
Green



Red	Blue	Green
1	0	0
0	1	0
0	0	1

Now each color is represented as a vector:

$$v_{\text{Red}} = [1, 0, 0]$$

$$v_{\text{Blue}} = [0, 1, 0]$$

$$v_{\text{Green}} = [0, 0, 1]$$

Why One-Hot Encoding is required?

If we used numbers like Red=1, Blue=2, Green=3, the model might think Green > Blue > Red, which isn't true. - **Eliminates ordinality**

Limitations of One-Hot Encoding

Vector dimension = number of words in vocabulary (e.g., 500,000)

- ✓ It is a large dimensional sparse vector representation.

Example.

The quick brown fox jumps over a lazy dog.

Vector representation of fox: [0,0,0,1,0,0,0,0,0]

- ✓ Also, it treats each word as independent of other word, even if both the words are together

$$\mathbf{v}_{notebook} = \mathbf{v}_{note} + \mathbf{v}_{book}$$

The vectors are orthogonal, and there is no natural notion of similarity between one-hot vectors!

Bag of Words

The **Bag of Words** model is a way to represent text data numerically so that algorithms can process it.

- Treats each document (sentence, paragraph, etc.) as a “bag” of words.
- Ignores grammar and word order.
- Only cares about the **frequency** (or presence) of words.

Bag of Words(BoW)

Example:

Sentence 1:I love NLP

Sentence 2: I love Deep Learning

Vocabulary

[I, love, NLP, Deep Learning]

	I	Love	NLP	Deep	Learning
Sentence 1	1	1	1	0	0
Sentence 2	1	1	0	1	1

Vectors for Sentence 1 and 2:

Sentence 1 : [1, 1, 1, 0,0]

Sentence 2 : [1, 1, 0, 1,1]

Limitation of Bag of Words

Sentence 1: Cat eats fish

Sentence 2: Fish eats cat

Vocabulary

[Cat, eats, fish]

	Cat	eats	fish
Sentence 1	1	1	1
Sentence 2	1	1	1

Sentence 1 : [1, 1, 1]

Sentence 2 : [1, 1, 1]



Word order information is lost

How to overcome limitations?

- The meaning of a word is its use in the language.
 - *How can we define the word meaning that encodes this notion of similarity?*
 - *Can we use shorter vectors?*
- One way to define "usage":
- words are defined by their environments (the words around them)
 - If A and B have almost identical environments we say that they are **synonyms**.
- Defining meaning as a point in space based on distribution
- Each word = a vector
- Similar words are "**nearby in semantic space**"
- We build this space automatically by seeing which words are **nearby in text**

